

# Geometric areas



- 1
- a 12 units<sup>2</sup>      b 21 units<sup>2</sup>      c 4 units<sup>2</sup>      d 12 units<sup>2</sup>

- 2
- a 18      b 72      c 104      d 38.5  
 e 18      f 80      g  $18\pi$       h  $9\pi$   
 i  $32\pi$       j  $18\pi$

- 3
- a
- i
- 
- ii 3 units<sup>2</sup>
- b
- i
- 
- ii  $\frac{51}{2}$  units<sup>2</sup>
- c
- i
- 
- ii 16 units<sup>2</sup>
- d
- i
- 
- ii 36 units<sup>2</sup>

## Approximate areas



4

- a An overestimation as the function is increasing on the given interval.
- b An underestimation as the function is decreasing on the given interval.

5

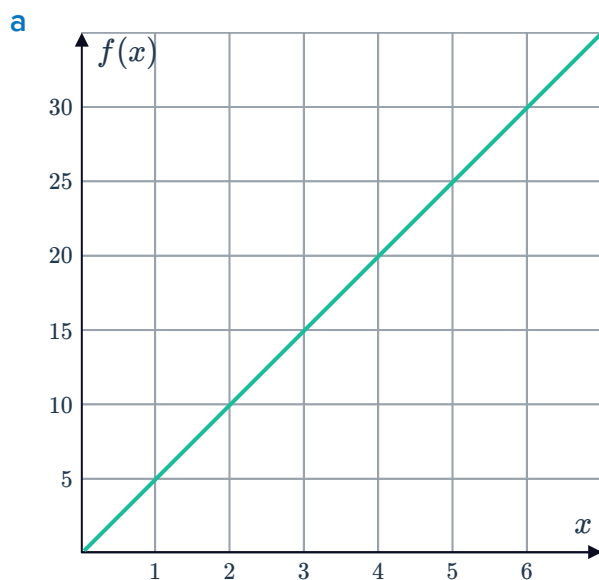
- a An underestimation as the function is increasing on the given interval.
- b An overestimation as the function is decreasing on the given interval.

6 The left endpoint approximation because the function is always increasing.

7

- a
  - i Overestimate
  - ii Underestimate
  - iii Underestimate
  - iv Overestimate
- b
  - i Underestimate
  - ii Overestimate
  - iii Overestimate
  - iv Underestimate

8

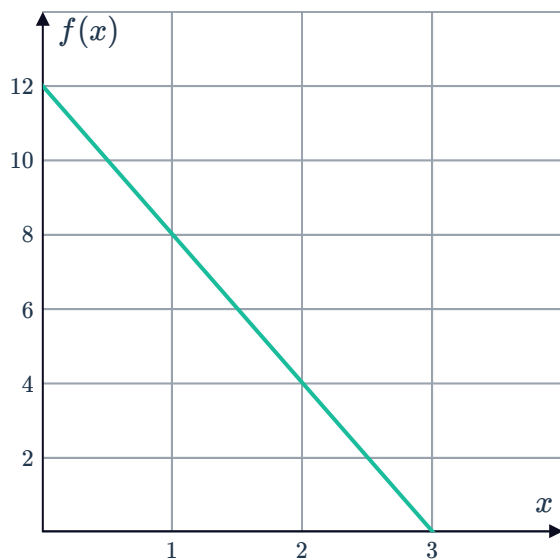


- b
- i 60 units<sup>2</sup>
  - ii 120 units<sup>2</sup>

- c
- i 75 units<sup>2</sup>
  - ii 105 units<sup>2</sup>
- d  $A = 90 \text{ units}^2$

9

a



b

i 24 units<sup>2</sup>ii 12 units<sup>2</sup>

c

i 21 units<sup>2</sup>ii 15 units<sup>2</sup>d 18 units<sup>2</sup>

10

a 62 units<sup>2</sup>b 50 units<sup>2</sup>11 336 units<sup>2</sup>12  $\frac{77}{60}$  units<sup>2</sup>

13

a 432 units<sup>2</sup>b 576 units<sup>2</sup>

## Use technology

14

a 30

b  $e^3 - 1$ 

c 36

15

a

| $n$                      | 5     | 10    | 100   | 1000  | 10 000 |
|--------------------------|-------|-------|-------|-------|--------|
| $A_L$ units <sup>2</sup> | 3.560 | 3.940 | 4.293 | 4.329 | 4.333  |

b  $\frac{13}{3}$  units<sup>2</sup>

16



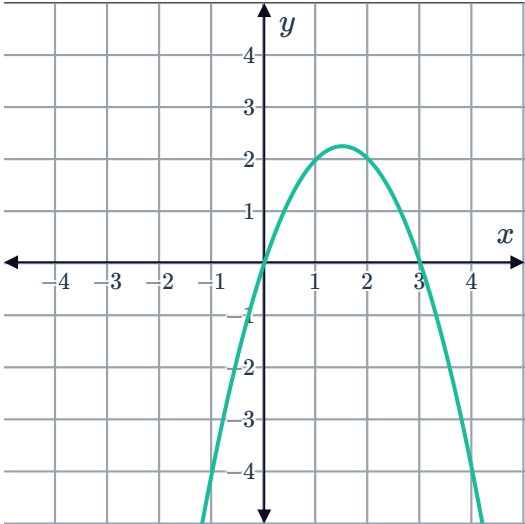
a

|                       |         |         |         |         |
|-----------------------|---------|---------|---------|---------|
| $n$                   | 5       | 10      | 100     | 1000    |
| $A_R \text{ units}^2$ | 2.63705 | 2.90452 | 3.12042 | 3.13956 |

b  $\pi \text{ units}^2$

17

a



b  $0 \leq x \leq 3$

c

|                       |        |        |        |        |        |
|-----------------------|--------|--------|--------|--------|--------|
| $n$                   | 5      | 10     | 100    | 1000   | 10 000 |
| $A_R \text{ units}^2$ | 4.3200 | 4.4550 | 4.4996 | 4.5000 | 4.5000 |

d  $\frac{9}{2} \text{ units}^2$